

SENIOR ENGINEER TO AI-NATIVE ENGINEER

From Writing Prompts to Architecting Resilient Autonomous Systems

1. Software 1.0 (Deterministic Code)

THE CLASSICAL BASELINE



Explicit Logic

Deterministic if/else statements
& rule engines



Algorithms

Static data structures & typed
compile-time models



Relational Data

Strict ACID schemas, foreign keys
& B-Tree indexes



Linear Pipelines

Synchronous request-response
APIs & microservices



Unit & E2E Tests

Deterministic assertions & static
boundary mocks

2. Software 2.0 (Statistical Machine Learning)

THE PROBABILISTIC SHIFT



Learned Weights

Gradient descent & high-
dimensional parameters



Transformers

Multi-head self-attention &
contextual embeddings



Vector Spaces

Semantic embeddings &
approximate nearest neighbors



Probabilistic Outputs

Logit distributions, sampling
strategies & top-p



Loss Benchmarks

Precision, recall, perplexity & ROC
curves

3. Software 3.0 (Agentic Systems & MCP)

THE SENIOR AI TARGET

LEARNING CURRICULUM & STAFF AI INTERVIEW FOCUS

Mastering the 20% of Deep Concepts That Govern 80% of Architectural Decisions

The 9 Core Learning Modules & Capstones

9 PRODUCTION PHASES • 6 LABS

PHASES 00 - 01

Foundations & Context

- BPE Tokenizer economics
- Attention & KV-Cache VRAM
- FSM logit masking grammars
- In-context learning & prompts

PHASES 02 - 03

RAG & MCP Tooling

- BM25 + pgvector Hybrid Search
- Reciprocal Rank Fusion (RRF)
- Cross-Encoder reranking
- MCP Client-Host-Server

PHASES 04 - 05

Agents & AI Security

- ReAct loops & Swarm mesh
- Anthropic 5 workflow patterns
- Dual-LLM privilege isolation
- Canary tokens & OWASP Top 10

PHASES 06 - 08

Evals, LLMOps & SDLC

- LLM-as-a-judge CI/CD gates
- OpenTelemetry spans & tracing
- Resilient AI API Gateways
- Semantic caching & failover

The 80/20 AI Architect Interview Triad

SENIOR & STAFF LEVEL

PILLAR 1

System Design Blueprints

- **Hybrid RAG:** BM25 + dense vectors + Cross-Encoder reranking with RBAC JWT isolation.
- **Resilient AI Gateway:** Multi-provider failover, semantic caching, token rate-limits.

PILLAR 2

Hardware Reality & Math

- **KV-Cache Sizing:** $2 \times 2 \times L \times H \times D \times S$ bytes per concurrent stream.
- **Prefill vs Decode:** Prefill is $O(N^2)$ compute-bound; Decode is $O(1)$ memory-bound.

PILLAR 3

Production Controls

- **Structured Outputs:** Outlines / Guidance FSM grammars for 100% schema validity.
- **Dual-LLM Quarantine:** Privileged executor vs untrusted parser separation.